HW 1

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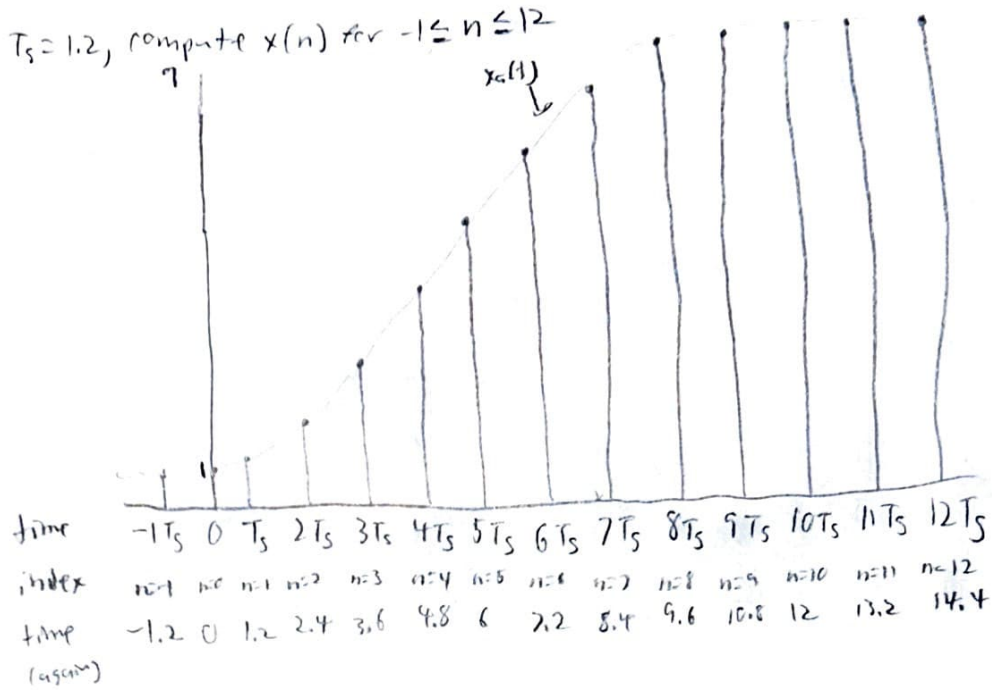
ECEN 133

1.

a. $x_a(t) = \begin{cases} 1 & t<1 \\ t & 1 \le t \le 9 \\ 9 & 9<t \end{cases}$ for $T_s = 1.2$, compute $x(n)$ for $-1 \le n \le 12$

(t)

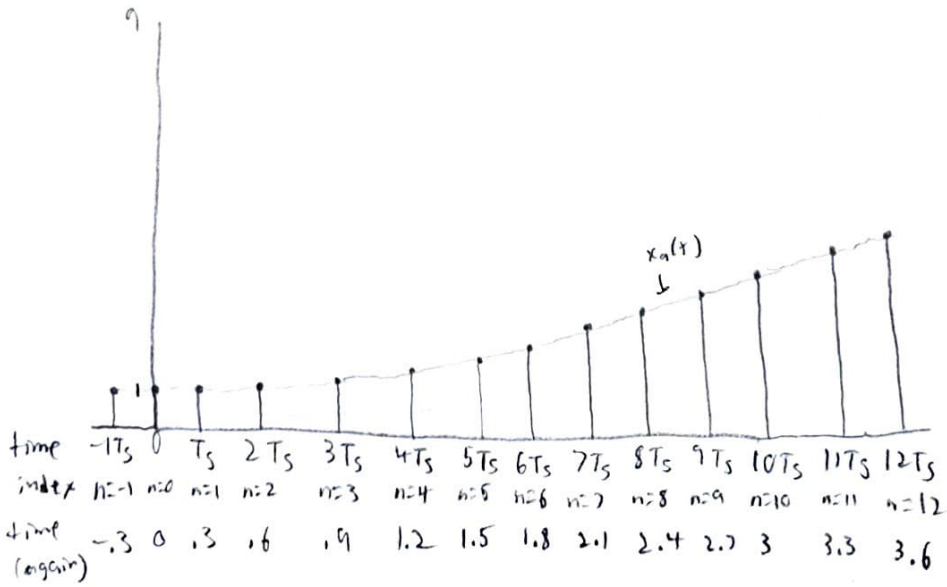
n	$n \cdot T_s$	$x_a(t)$
-1	-1.2	1
0	0	1
1	1.2	1.2
2	2.4	2.4
3	3.6	3.6
4	4.8	4.8
5	6	6
6	7.2	7.2
7	8.4	8.4
8	9.6	9
9	10.8	9
10	12	9
11	13.2	9
12	14.4	9



b.

(t)

n	$n \cdot T_s$	$x_a(t)$
-1	-.3	1
0	0	1
1	.3	1
2	.6	1
3	.9	1
4	1.2	1.2
5	1.5	1.5
6	1.8	1.8
7	2.1	2.1
8	2.4	2.4
9	2.7	2.7
10	3	3
11	3.3	3.3
12	3.6	3.6



2.

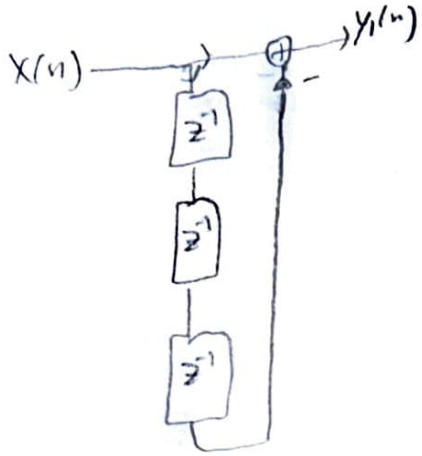
$x_a(t) = A\cos(2\pi(200)t) + B\sin(2\pi(200)t)$

for $F_T = \begin{cases} 1600, 800 \\ 400, 200 \end{cases}$ do a-f First 12: $n = 0, 1, \ldots, 11$

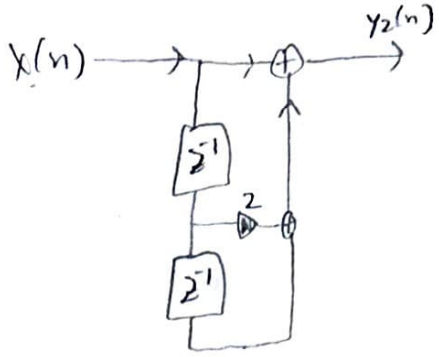
	$F_T = 1600$	$F_T = 800$	$F_T = 400$	$F_T = 200$
a.	$T_S = \frac{1}{F_T} = \frac{1}{1600}$ sec/sample	$\frac{1}{800}$ sec/sample	$\frac{1}{400}$ sec/sample	$\frac{1}{200}$ sec/sample
b.	$1600(.2) = 320$ samples	$800(.2) = 160$ samples	$400(.2) = 80$ samples	$200(.2) = 40$ samples
c.	8 samples long	4 samples	2 samples	1 sample
d.	$v_0 = \frac{f_0}{F_T} = \frac{200}{1600} = .125$ cycles/sample	$\frac{200}{800} = .25$ cycles/sample	$\frac{200}{400} = .5$ cycles/sample	$\frac{200}{200} = 1$ cycle/sample
e.	8 samples/cycle	4 sam/cycle	2 sam/cycle	1 sam/cycle
f.	$t = nT_S$ $x_a(t) = A\cos(\frac{\pi n}{4}) + B\sin(\pi n/4)$ $n=0$: A $n=1$: $\frac{\sqrt{2}}{2}(A+B)$ $n=2$: B $n=3$: $\frac{\sqrt{2}}{2}(B-A)$ $n=4$: $-A$ $n=5$: $-\frac{\sqrt{2}}{2}(A+B)$ $n=6$: $-B$ $n=7$: $\frac{\sqrt{2}}{2}(A-B)$ $n=8$: A $n=9$: $\frac{\sqrt{2}}{2}(A+B)$ $n=10$: B $n=11$: $\frac{\sqrt{2}}{2}(B-A)$	$x_a(t) = A\cos(\frac{\pi n}{2}) + B\sin(\frac{\pi n}{2})$ $n=0$: A $n=1$: B $n=2$: $-A$ $n=3$: $-B$ $n=4$: A $n=5$: B $n=6$: $-A$ $n=7$: $-B$ $n=8$: A $n=9$: B $n=10$: $-A$ $n=11$: $-B$	$x_a(t) = A\cos(\pi n) + B\sin(\pi n)$ $n=0$: A $n=1$: $-A$ $n=2$: A $n=3$: $-A$ $n=4$: A $n=5$: $-A$ $n=6$: A $n=7$: $-A$ $n=8$: A $n=9$: $-A$ $n=10$: A $n=11$: $-A$	$x_a(t) = A\cos(2\pi n) + B\sin(2\pi n)$ for $n = 0, 1, \ldots, 11$: $x_a = A$ (all A)

3.

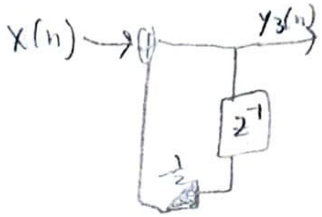
a. $y_1(n) = x(n) - x(n-3)$



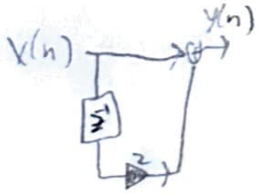
b. $y_2(n) = x(n) + 2x(n-1) + y(n-2)$



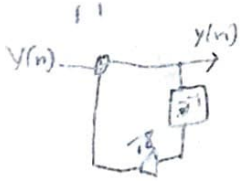
c. $y_3(n) = x(n) - \frac{1}{2} y_3(n-1)$



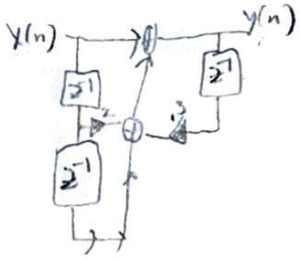
4. i



ii



iii



b. $y(n) = x(n) + 2(x(n-1))$

$y(n) = x(n) - .8y(n-1)$

$y(n) = x(n) + 2x(n-1) + x(n-2) + .3y(n-1)$

a. the adder sums $x(n)$ and $2x(n-1)$

the adder sums $x(n)$ and $-.8y(n-1)$

the low adder sums $2x(n-1)$, $x(n-2)$, and $.3y(n-1)$

the top adder sums the prev result and $x(n)$.